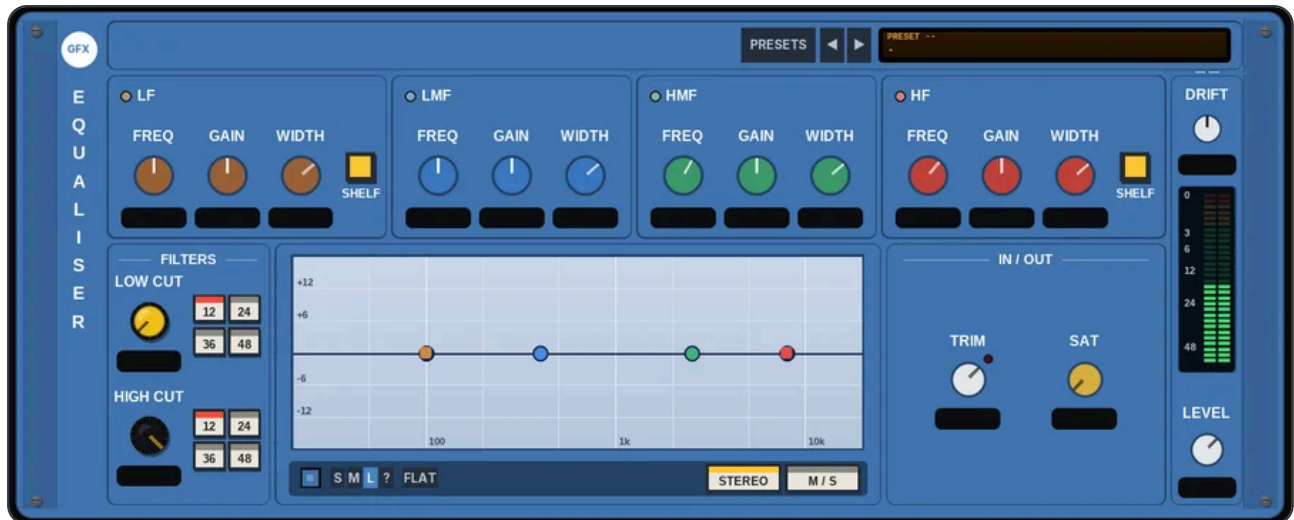


Equaliser

A musical four-band parametric with overlapping bands, notches, shelves, low and high cuts, and mid / side.



Overview

Equaliser is a four-band parametric in the spirit of the classic 70s broadcast parametrics. The four bands - LF, LMF, HMF and HF - have overlapping ranges, so two bands can be stacked on the same area. The width is 'proportional': small boosts and cuts are gentle and broad, and the band gets narrower as you push the gain further. The last part of each GAIN knob's cut dives into a deep notch, down to -40 dB, for removing hum or a single ringing frequency.

LF and HF switch between a bell and a shelf. Separate low-cut and high-cut filters, each with a choice of four slopes, sit in front of the bands. In M / S mode each band can work on both, the mid only or the sides only. You can also work straight on the curve: drag a band's dot to set its frequency and gain.

Reach for it for

- Broad, musical tone shaping on vocals, instruments and buses
- Notching out hum or a single ringing frequency
- Cleaning up the lows and highs with the LOW CUT and HIGH CUT filters
- Mid / side work: tightening the low end of the sides, adding air to the width
- Lo-fi telephone and radio sounds

Controls

The bands: LF, LMF, HMF, HF

Each band has its own colour (brown, blue, green and red), used for its knob caps, its dot and its shaded curve on the display.

FREQ	The band's frequency. LF: 20 - 500 Hz (100 Hz by default). LMF: 80 Hz - 2 kHz (400 Hz). HMF: 400 Hz - 8 kHz (2.5 kHz). HF: 1.5 - 20 kHz (8 kHz). The ranges overlap, so bands can be stacked.
GAIN	Boost or cut, -12 to +16 dB in the normal range. Past -12 dB the knob dives into a notch: the last quarter of the cut goes down to -40 dB (shown as NOTCH).
WIDTH	The band's width in octaves, 0.05 to 4 (1 by default). Small values are narrow; large values are broad. Bells get narrower anyway as the gain is pushed. On a shelf, WIDTH sets how gentle the slope is.
SHELF (LF and HF only)	Switches the band between a bell and a shelf. Both start as shelves.
ST / MID / SIDE	Shown in M / S mode only. What the band works on: ST both, MID the centre (what is common to both sides), SIDE the sides (the difference).

FILTERS

LOW CUT	A high-pass filter, from OFF (fully left, 20 Hz) up to 1 kHz.
HIGH CUT	A low-pass filter, from 2 kHz up to OFF (fully right, 20 kHz).
12 / 24 / 36 / 48	The slope of each filter in dB per octave, next to its knob. 12 is gentle; 48 is very steep. The filters have no bump at the corner.

The display

The curve covers 20 Hz to 20 kHz and plus or minus 18 dB. Each band that is boosting or cutting is shaded in its own colour, with the total curve drawn on top. In M / S mode, when any band works on the mid or side only, a second purple curve shows the side and the labels MID and SIDE appear.

Band dots	Drag a band's dot to set its frequency (left and right) and gain (up and down). Shelves show a square dot.
Mouse wheel on a dot	Changes the band's WIDTH.
STEREO / M / S	Under the curve. STEREO works on left and right; M / S lets each band work on the mid or the sides.

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	Click the TRIM knob (without moving it) to switch the guard; its light sits at the knob's upper right and is lit while it is on. Off by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The light glows brighter while it is catching peaks.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.

Drift

A slow random wander, like an unsteady power supply: the band frequencies and gains move together. Bands set flat stay flat.

DRIFT	How much drift, 0 to 100%. 50% (the default) is the standard amount, 100% is twice that, 0 switches it off. More drift also makes the wander a little faster.
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Output

Meter	Output level, left and right.
LEVEL	Output level, -24 to +12 dB.

The preset row and colour strip

PRESETS	Opens the list of factory presets (a click on the display does the same). Choosing one sets the bands, filters and STEREO / M / S; drift, output level and the IN / OUT section are left as they are. The last entry resets everything to the defaults.
◀ / ▶	Step to the previous or next preset.
Colour swatch	Changes the faceplate colour: GREEN, BLUE or one of the GFX themes.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Next to the S / M / L size keys. Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
HW / FLAT	Switches between the hardware-style panel and the flat look.

Presets

Vocal Presence	Low cut at 80 Hz, a little less low end and mud, +3 dB at 3 kHz and a +3 dB air shelf at 10 kHz.
Kick Weight	+4 dB bell at 60 Hz, -4 dB at 350 Hz, +3 dB click at 3.5 kHz; low cut at 30 Hz.
Bass Clarity	A little low shelf, -3 dB at 250 Hz, +2 dB at 800 Hz for definition, a slightly softer top.
Acoustic Sparkle	Lows and boxiness trimmed, +2 dB at 5 kHz and a +4 dB

	shelf at 12 kHz; low cut at 70 Hz.
Mix Air	A broad +3 dB shelf at 15 kHz, low cut at 25 Hz.
Mud Cut	-4 dB at 300 Hz, low cut at 50 Hz.
Telephone	Low cut at 400 Hz and high cut at 3.5 kHz (24 dB per octave) with +6 dB at 1.5 kHz.
De-Harsh	A narrow -4 dB cut at 3.2 kHz and -3 dB at 7 kHz.
Hum Notch 50 Hz	Deep, narrow notches at 50 Hz and 100 Hz.
M/S Wide Air	M / S mode, all on the sides: a -12 dB low shelf at 120 Hz, a little less mud and a +4 dB air shelf at 10 kHz.
M/S Centre Focus	M / S mode: a -10 dB low shelf at 150 Hz on the sides and +2 dB presence at 2 kHz on the mid.
Reset all to defaults	Every control back to its starting value.

Tips

- Because the width is proportional, small moves stay broad and musical - you rarely need to touch WIDTH for gentle tone shaping.
- To find a ringing frequency, boost a narrow band, sweep it until the ring jumps out, then turn GAIN right down into the notch.
- Stack two bands on the same area (the ranges overlap) for a bigger or more complex boost or cut.
- In M / S mode, try a low shelf cut on SIDE to keep the low end centred and solid.

File: gfx-equaliser.jsfx